
Chapter 4: Data Handling and Presentation

Worksheet 4: Organising and Representing Data

Learning Objective: Students will collect and organise data using tally marks and frequency tables; represent data using bar graphs and pictographs; calculate mean, median, and mode; interpret graphical data.

Worked Example

Scores of 7 students: 8, 5, 7, 8, 6, 8, 5.

$$\text{Mean} = \frac{8+5+7+8+6+8+5}{7} = \frac{47}{7} \approx 6.7$$

Mode = 8 (appears most often)

Median: Arrange in order: 5, 5, 6, 7, 8, 8, 8. Middle value = 7

Questions 1–5: Easy / Recall

Q1. What is the mean (average) of: 4, 8, 6, 10, 2?

Q2. Find the mode of: 3, 7, 3, 9, 3, 7, 5.

Q3. Find the median of: 12, 5, 9, 3, 7.

Q4. How many tally marks represent the number 17?

Q5. What type of graph uses symbols/pictures to represent data?

Questions 6–10: Basic Application

Q6. The marks of 8 students are: 45, 62, 58, 71, 45, 89, 62, 45. Find mean, mode, and median.

Q7. Make a frequency table for: Red, Blue, Red, Green, Blue, Red, Yellow, Blue, Red, Green.

Q8. A bar graph shows runs scored by a cricket team: Mon=45, Tue=60, Wed=30, Thu=75, Fri=50. Find the average daily run.

Q9. In a survey, 30 students chose their favourite subject. Maths: 12, Science: 8, English: 6, Hindi: 4. Draw a frequency table and state which subject is most popular.

Q10. If the mean of 5 numbers is 20, what is their sum?

Questions 11–15: Practice and Skill Building

Q11. The ages of 10 players on a cricket team: 18, 22, 19, 21, 20, 23, 18, 24, 20, 20. Find mean, mode, and median.

Q12. Heights of 6 plants (cm): 15, 22, 18, 22, 25, 18. Find the mean and both modes.

- Q13.** The mean of 6 numbers is 15. If five of them are 12, 18, 14, 16, 10, find the sixth number.
- Q14.** Data: 23, 28, 19, 35, 28, 41, 19, 28, 35. Find mean, mode, median, and range (max – min).
- Q15.** Create a tally table and bar graph description for: Class scores between 0–20: 6 students, 21–40: 8 students, 41–60: 14 students, 61–80: 10 students, 81–100: 2 students.

Questions 16–18: Higher Order Thinking

- Q16.** The mean score of 20 students in a test is 72. When the teacher re-checks, one student's score is corrected from 48 to 84. What is the new mean?
- Q17.** The median of 7 values is 25. The values are 18, 21, x , 25, 30, 32, 40 (already sorted). Find x . Can there be more than one value of x ?
- Q18.** Raman says "the mean is always the best measure of data." Priya says "not always; sometimes the mode or median is better." Who is correct and why? Give an example to support the better choice.

Questions 19–20: Real-Life Applications

Q19. Rainfall (mm) in a city for 6 months: Jan=10, Feb=5, Mar=12, Apr=30, May=80, Jun=150. Find the mean monthly rainfall. In which months was rainfall above average? Make a rough bar graph. (Teacher Verification Required for graph)

Q20. Survey your class: How do you come to school? (Walk / Cycle / Bus / Auto). Collect data, make a tally table, find the mode (most common mode of transport), and draw a pictograph. (Teacher Verification Required)

Reflection Question: Can a dataset have more than one mode? Can it have no mode? Give examples.